

Active Morphological Probing: Terrain-Aware Locomotion Without Depth Perception

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Abstract: Wheeled-legged robots gives the versatility of legged locomotion on unstructured terrain and the speed and efficiency of wheeled movement on flat ground. The existing robot design actively uses all degrees of freedom to move across different terrains incorporates by adding wheel to the feet of the robots, which adds four extra degrees of freedom at the feet of the quadruped robots that makes it less power efficient. In this paper, we present an open source morphologically adaptive robot that actively reconfigures between wheeled and legged locomotion to move effortlessly between varied terrains.

but typically require depth cameras and complex algorithms to determine when to reconfigure. We argue that robot design choices directly determine algorithmic complexity a robot built with learnability in mind can reduce a hard perception-and-control problem to a simple learned policy. We present AWM (All-Wheel-Morph), a low-cost open-source wheeled-legged robot where each wheel extends into a leg, designed to make morphological adaptation learnable: actuators with built-in torque observability, a continuous morphology range, and sim-to-real transferable dynamics. Leveraging torque feedback natively available from AWM’s actuators, we propose **Active Morphological Probing** a GRU policy that physically senses terrain by extending legs and reading contact torque, adapting morphology without any depth camera. Without exteroception, our policy matches camera-equipped baselines in terrain traversal while a fixed-morphology wheels-only policy plateaus at half the curriculum difficulty regardless of training duration, confirming that adaptive morphology is the key enabler. Our approach requires no teacher-student distillation, no privileged training information, and deploys as a single network. We fully open-source AWM including hardware design, bill of materials, simulation assets, training code, and trained policies so that anyone can build, train, and deploy a morphologically adaptive robot with off-the-shelf components under \$2,500.

Keywords: morphological adaptation, legged locomotion, proprioception, torque sensing, sim-to-real

1 Introduction

Traversing unstructured terrain stairs, curbs, rough ground remains an open problem for wheeled robots. The dominant solution couples a depth camera with a learned policy that reads a height map of the upcoming surface and decides how to reconfigure or accelerate [1, 2]. This works, but it couples the robot’s intelligence to a sensor that adds cost, weight, and failure modes: cameras saturate in bright light, fail on reflective or textureless surfaces, and require careful extrinsic calibration that drifts over time.

We take a different starting point. Rather than asking what sensor can tell the robot about terrain, we ask what physical design makes terrain information natively available. Our answer is morphological

39 probing: a robot that extends its legs into contact with the surface and reads the resulting joint
40 torques. Stairs feel different from flat ground through torque alone — a stair edge loads the leg
41 abruptly, producing a distinctive current spike that a GRU can learn to recognize over time. No
42 depth image required.

43 This idea only works if the robot is designed for it. A leg with a high-ratio gearbox in position-
44 control mode cannot sense torque: gear friction holds most of the load, and motor current does not
45 reflect contact force. We present AWM (All-Wheel-Morph), a wheeled-legged platform specifically
46 designed to make torque sensing reliable: current-based position control (operating mode 5) on all
47 leg actuators, calibrated to within 2% across 18 trials, producing torque estimates that map directly
48 to the simulation’s observation space without additional processing.

49 We train a GRU policy over a 25-dimensional proprioceptive observation that includes four normal-
50 ized leg torques but no exteroception. The result is **Active Morphological Probing (AMP)**: the
51 policy learns to extend legs as terrain probes, read the torque response, and adapt morphology ac-
52 cordingly. In simulation on a staircase curriculum, AMP matches the terrain traversal performance
53 of a camera-equipped baseline (GRU + terrain_scan: 0.72 vs. AMP: 0.72 ± 0.03), while a wheels-
54 only policy with identical training plateaus at 0.42, confirming that adaptive morphology — not just
55 a better policy — is the key enabler.

56 Our contributions are:

- 57 1. **AWM**: an open-source wheeled-legged robot under \$2,500 with calibrated torque sensing
58 via current-based position control, designed for sim-to-real transfer.
- 59 2. **Active Morphological Probing**: a GRU policy that physically senses terrain through leg
60 contact torques, requiring no depth camera, teacher network, or privileged training signal.
- 61 3. Simulation experiments showing AMP matches camera-equipped baselines and dramat-
62 ically outperforms fixed-morphology wheels-only locomotion. [TODO: add: real-robot
63 sim-to-real validation]
- 64 4. Full open-source release: CAD, bill of materials, simulation assets, training code, and
65 trained weights.

66 2 Related Work

67 **Terrain-aware locomotion with exteroception.** Miki et al. [2] show that an attention-based re-
68 current encoder fusing proprioception with a depth-camera height map enables robust locomotion
69 in unstructured outdoor environments. Agarwal et al. [3] achieve stair and gap traversal using only
70 a front-facing depth camera in an end-to-end policy. Lee et al. [4] extend perceptive locomotion
71 to wheeled-legged ANYmal, validating km-scale navigation in urban environments with privileged
72 height-map training. All of these require exteroceptive sensors at deployment; AMP matches their
73 terrain performance without any depth sensor by using leg torques instead.

74 **Proprioceptive and blind locomotion.** Lee et al. [1] showed that an LSTM over propriocep-
75 tive history enables challenging-terrain quadruped locomotion without any camera. RMA [5] ex-
76 tends this with a two-phase approach: a base policy trained with privileged environment informa-
77 tion and an adaptation module that infers those factors from proprioceptive history at deployment.
78 DreamWaQ [6] learns an implicit terrain representation through a latent state estimator without
79 exteroception. Ji et al. [7] use a GRU-based internal model with joint encoders and IMU only, es-
80 timating terrain state from the robot’s own dynamic response. Margolis et al. [8] achieve 3.9 m/s
81 blind locomotion on MIT Mini Cheetah with a gait-conditioned proprioceptive policy. All of these
82 reactive approaches detect terrain *after* contact; AMP deliberately extends legs *before* committing
83 to traversal, gathering terrain information actively.

84 **Morphologically adaptive and wheeled-legged robots.** Wheeled-legged platforms combine
85 rolling efficiency with stepping capability for versatile terrain coverage [9, 10, 11]. Rudin et al.

[TODO: Figure: robot photo (real) + CAD render side by side, with leg extended and retracted positions labeled]

Figure 1: AWM (All-Wheel-Morph). Each of the four wheels integrates a leg actuator whose rotation extends the wheel hub into ground contact. At left: legs retracted (wheel mode). At right: legs extended (probing/stair mode). All hardware is open-sourced under \$2,500.

[12] demonstrate blind stair climbing on the wheeled-legged Ascento using asymmetric actor-critic RL, achieving sim-to-real transfer without exteroception. These works treat morphology as fixed or switch between discrete modes via trajectory optimization. AWM learns a *continuous* morphological adaptation policy driven entirely by sensory feedback, without any explicit mode-switching logic.

Curriculum learning for locomotion RL. Rudin et al. [13] established the massively parallel terrain curriculum paradigm that Isaac Lab descends from, training locomotion policies in minutes on thousands of parallel environments with automatic difficulty progression. Walk These Ways [14] extends this with gait command conditioning for behavioral generalization across locomotion styles. AWM’s terrain curriculum follows this design directly.

Joint torques and contact sensing. Joint torque sensing is used for contact detection, impedance control, and collision identification in manipulation [15], and for foot contact detection in legged locomotion. Higuera et al. [16] use foot-mounted force probes for terrain traversability assessment in a planning module separate from the locomotion policy. To our knowledge, no prior work feeds raw joint torques as a direct observation into a locomotion RL policy. We show that a GRU over 4-dimensional leg torque history can replace a 35-dimensional terrain height map for morphological adaptation decisions.

3 The AWM Platform

3.1 Mechanical Design

AWM is a four-wheeled robot in which each wheel hub is coupled to a rotary leg actuator. Rotating the actuator extends the wheel outward and downward, transitioning from pure rolling to a hybrid wheel-leg contact. The morphology range is continuous: from fully retracted (wheels contact ground, legs clear) to fully extended (legs contact ground, providing torque measurement). This design choice — continuous rather than discrete morphology — is deliberate: it gives the policy a smooth action space and allows partial extension as a probing strategy without committing to a full stance change.

Wheels are driven by XM430-W210-R actuators (RS-485 bus, velocity control). Legs are driven by XM430-W350-T actuators (TTL bus, current-based position control, stall torque 4.1 Nm at 12 V). Four slip rings (6-wire, 10 A rated) provide power and signal continuity through leg rotation. The robot is designed for straightforward assembly from off-the-shelf components; the full bill of materials is provided in the supplementary materials.

3.2 Torque Sensing via Actuation Mode

The key hardware insight is that torque sensing quality depends critically on actuator operating mode. In standard position control (mode 3), the high gear ratio (353:1) of the XM430-W350-T causes gear friction to hold most external load — Present Current reflects the motor’s effort to maintain position against friction, not against terrain. We switch all leg actuators to **current-based position control (mode 5)**: the motor actively maintains a goal position by pushing against terrain, and Present Current directly reflects terrain reaction force.

We calibrate the torque constant empirically: a rigid test arm at three distances (40, 80, 120 mm) with two known loads (500 g, 1 kg) across 18 trials yields $k_\tau = 3.83 \text{ Nm/A}$ (std < 0.12 Nm/A across

all conditions). The conversion is:

$$\tau_{Nm} = I_{LSB} \times 2.69 \times 10^{-3} \times 3.83 \quad (1)$$

where I_{LSB} is the raw Present Current register value and 2.69 mA/LSB is the unit from the actuator datasheet. This pipeline runs at 60 Hz on the Jetson Orin Nano Super, matching the simulation policy rate.

3.3 Open-Source Release

AWM is fully open-sourced: CAD files, bill of materials, simulation assets (Isaac Lab USD), training code, and trained policy weights are released at [TODO: URL]. The complete platform can be assembled from off-the-shelf components for under \$2,500, enabling any lab to replicate our experiments.

4 Active Morphological Probing

4.1 Problem Formulation

We formulate locomotion as a Markov Decision Process (MDP) with state s_t , action a_t , and reward r_t . The robot follows a commanded velocity (v_x, ω_z) . The policy π_θ maps an observation history to actions: wheel velocities for the four drive motors and leg position targets for the four leg actuators.

4.2 Observation Space

The AMP policy receives a 25-dimensional proprioceptive observation with no exteroceptive signal:

Observation	Dims	Source
Commanded velocity (v_x, ω_z)	2	Joystick / planner
Base linear velocity v_x	1	Wheel odometry
Base angular velocity ω_z	1	IMU (gyro z)
Wheel velocities	4	Drive motor encoders
Leg positions	4	Leg motor encoders
Previous leg actions	4	Policy memory
Projected gravity	3	IMU (accelerometer)
Progress-slip history	2	Computed
Leg torques	4	Mode-5 current (Eq. 1)
Total	25	

Table 1: AMP observation space. Leg torques (bold) replace the 35-dimensional terrain_scan from the camera baseline, reducing observation size by 31 dimensions while enabling terrain sensing through physical contact.

Torque sensing via PD loading. The leg torque signal arises from a deliberate interaction between the policy and the PD controller. At each timestep, the policy outputs a leg position target q^{des} , and the actuator applies torque according to:

$$\tau = k_p(q^{\text{des}} - q) + k_d(0 - \dot{q}) \quad (2)$$

On flat ground the leg reaches q^{des} easily and τ remains small. When the leg contacts a stair edge, terrain physically prevents q from reaching q^{des} — the position error $(q^{\text{des}} - q)$ stays large, the motor draws proportionally more current, and τ spikes. This spike enters x_{t+1} as the leg torque observation. Critically, the policy *causes* this signal by commanding a target the terrain cannot satisfy — probing terrain stiffness rather than passively measuring contact. The GRU integrates this torque history over ≈ 0.8 s to distinguish terrain types: a stair produces a sustained asymmetric spike pattern across the four legs, while flat ground produces near-zero symmetric readings.

Leg torques are normalized by the actuator effort limit (2.70 Nm) and clamped to $[-2, 2]$ to handle transient impact spikes. Gaussian noise ($\text{std} = 0.05$) is added during training for sim-to-real robustness.

The camera baseline (GRU + terrain_scan) uses a 56-dimensional observation including a 7×5 height grid from a ray-caster sensor (in simulation) or ZED Mini depth camera (on the real robot), with matched Gaussian noise ($\text{std} = 0.02$, 2 cm depth noise model).

4.3 Policy Architecture

Both AMP and the camera baseline use the same GRU architecture: a single-layer GRU with 256 hidden units followed by a two-layer MLP actor head (hidden dims [256, 128], ELU activations). The GRU processes the full observation at each timestep and maintains hidden state across the episode, allowing the policy to integrate torque history for terrain identification. The recurrent rollout length is 48 steps (≈ 0.8 s at 60 Hz). A separate critic network shares the same structure. Action noise is initialized at $\sigma = 0.3$ (scalar, not state-dependent).

4.4 Training

We train using Proximal Policy Optimization (PPO) [17] in Isaac Lab [18] with 1024 parallel environments. The terrain curriculum consists of flat ground (50%) and random uniform staircases (50%, step height 0.05–0.20 m, step width 0.25–0.35 m). The curriculum difficulty (terrain_levels, $\in [0, 1]$) advances when the mean episode reward exceeds a threshold, and the metric at convergence is our primary performance measure.

The reward function encourages velocity tracking and penalizes tilt, wheel slip, and excessive joint velocities:

$$r_t = r_{\text{vel}} + r_{\text{yaw}} + r_{\text{leg}} - c_{\text{tilt}} - c_{\text{slip}} - c_{\text{joint}} - c_{\text{action}} \quad (3)$$

A morphology-specific bonus r_{leg} rewards leg extension on rough terrain and penalizes retracted legs when stuck, encouraging the emergent probing behavior. All policies are trained for 30,000 iterations ($\approx 1.47 \times 10^9$ environment steps).

5 Experiments

We evaluate three questions: (1) Does torque sensing match camera performance in simulation? (2) Is the morphology specifically contributing, or just having legs? (3) Does the policy transfer to the real robot?

5.1 Simulation Comparison

We compare four conditions trained on the same curriculum for the same number of iterations:

Method	Architecture	Obs dims	terrain_levels	Seeds
WheelsOnly	MLP	21	0.42	1
MLP + terrain_scan	MLP	56	0.68 ± 0.02	3
GRU + terrain_scan	GRU	56	0.72	1
AMP (ours)	GRU	25	0.72 ± 0.03	4

Table 2: Terrain curriculum level at convergence (mean $\pm$ std across seeds, last 10 evaluations per run). AMP matches the GRU camera baseline while using 31 fewer observation dimensions and no depth sensor.

Three findings stand out. First, AMP (0.72 ± 0.03) matches GRU + terrain_scan (0.72) on the apples-to-apples comparison: same architecture, same training, only the sensing modality differs. Second, morphology is essential: WheelsOnly plateaus at 0.42 regardless of training duration, a gap of 0.30

185 terrain levels that no policy improvement can close. Third, the recurrent architecture matters: MLP
 186 + terrain_scan (0.68) underperforms GRU + terrain_scan (0.72), confirming that temporal integration
 187 of terrain signals — whether from depth or torque — is important.

188 5.2 Ablation: What Does Torque Sensing Contribute?

189 [TODO: Run: GRU + legs extended fixed (no torque sensing) condition. This isolates whether
 190 torque specifically is doing the work vs. just having legs available. Expected result: fixed-leg con-
 191 dition underperforms AMP, showing torque information — not just morphology presence — drives
 192 the gain over WheelsOnly.]

Method	Legs	terrain_levels
WheelsOnly	Retracted (fixed)	0.42
GRU + fixed legs (no torque)	Extended (fixed)	[TODO: X.XX]
AMP (ours)	Adaptive + torque	0.72 ± 0.03

Table 3: Ablation isolating the contribution of torque sensing. Fixed-legs condition has legs ex-
 tended but not torque-sensing: it cannot adapt morphology and receives no torque input.

193 5.3 Torque Signatures as Terrain Encoding

194 [TODO: Figure: real robot leg torque traces over time on (a) flat ground, (b) stair approach and
 195 climb. Show that stairs produce a characteristic torque spike pattern that is temporally structured
 196 and repeatable across trials. This is the physical justification for why a GRU over torque history can
 197 substitute for a depth image.]

198 To understand *why* AMP works, Figure [TODO: X] shows leg torque traces from the real robot
 199 traversing flat ground and a three-step staircase. [TODO: Write 2–3 sentences describing the pattern
 200 after figure is available.]

201 5.4 Sim-to-Real Transfer

202 [TODO: Run both AMP and GRU + terrain_scan on the real robot. Report terrain_levels using the
 203 same metric as simulation (stair step height curriculum). Target: within 10% of sim performance for
 204 both conditions. Use ZED Mini IMU for both (same hardware), ZED Mini depth for terrain_scan
 205 condition only.]

Method	Sim	Real
GRU + terrain_scan	0.72	[TODO: X.XX]
AMP (ours)	0.72 ± 0.03	[TODO: X.XX]

Table 4: Sim-to-real transfer. Both policies use identical hardware (ZED Mini for IMU); terrain_scan
 additionally uses ZED Mini depth. The clean hardware match isolates the sensing modality as the
 only difference between conditions.

206 5.5 Failure Analysis

207 [TODO: After real-robot experiments: document conditions where AMP fails. Likely candidates:
 208 (1) terrain step height exceeding leg range, (2) slippery surfaces where torque ambiguously reflects
 209 slip vs. terrain resistance, (3) rapid step sequences that exceed GRU temporal integration window.]

210 6 Conclusion

211 We presented Active Morphological Probing, an approach that uses a robot’s own leg actuators as
 212 terrain sensors. By switching leg actuators to current-based position control and feeding normalized

213 torque signals into a GRU policy, we match the terrain traversal performance of a depth-camera-
214 equipped baseline without any exteroceptive sensor. The key insight is that robot design determines
215 sensing capability: the same actuators that create morphological adaptability also provide terrain
216 information, at no additional hardware cost.

217 The performance gap between WheelsOnly (0.42) and AMP (0.72) demonstrates that adaptive mor-
218 phology is the primary enabler of stair climbing performance — a gap that training time alone
219 cannot close. We release AWM, the full hardware and software stack, to make this a reproducible
220 community benchmark for morphologically adaptive locomotion.

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